Measurements of ultrasound properties in polyacryl and PVC

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The velocity of ultrasound waves in polyacryl was determined to be $c_{\text{poly, 1MHz}} = (2626.2 \pm 6.5)\,\text{m/s}$ for $f = 1\,\text{MHz}$ and $c_{\text{poly, 2MHz}} = (2655.3 \pm 6.7)\,\text{m/s}$ for $f = 2\,\text{MHz}$. The sound velocity in water was examined to be $c_{\text{water, 2MHz}} = (1478 \pm 14)\,\text{m/s}$. It was captured a B-mode image of a polyacrylic test block with boreholes at different positions. Additionally, these hole-distances from the upper edge were measured with the sound reflection method and compared with a caliper measurement. The resolution power of ultrasound was examined for $f_1 = 1\,\text{MHz}$ and $f_2 = 2\,\text{MHz}$. Resolving two close neighboring boreholes, the full width at half maximum for f_2 was determined to be $FWHM = (1.21 \pm 0.14)\,\mu\text{s}$ and thus f_2 exhibited the greater resolving power. The absorption coefficient for PVC was determined with both frequencies to be $\rho(f_1) = (0.093 \pm 0.017)\,1/\text{cm}$ and $\rho(f_2) = (0.53 \pm 0.13)\,1/\text{cm}$.

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1 Introduction

Moving ultrasound images of babies in the womb are nowadays familiar to most members of the public through either personal experience of antenatal scanning or via television. Modern ultrasound systems do much more than produce images of unborn babies, however. Modern ultrasound systems are able to make detailed measurements of blood movements in blood vessels and tissues, visualize moving structures in 3D [1], and make measurements related to the stiffness of tissues.
In this experiment, ultrasound was used to determine the velocity of sound and the absorption coefficient in solids. Also, the two dimensional structure of a polyacrylic brick was explored by examining reflection and transmission of ultrasound and a B-mode image was recorded.

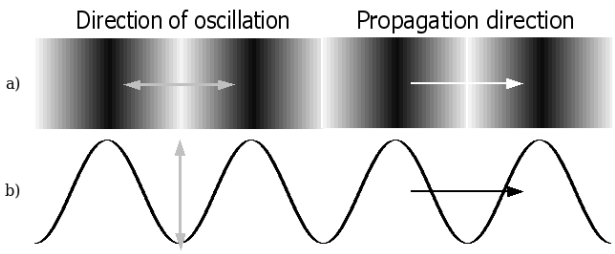

Figure 1: Illustrated are the two different waveform. a) shows a longitudinal wave, b) shows a transverse wave, where the atomic oscillations are orthogonal to the propagation direction. A real wave propagation through a 3D-lattice consists of one longitudinal and two transverse modes, which are commonly described in terms of particles - the so called phonons. Figure Adapted from [2].

2 Theory and experimental considerations

In solid material, the atoms are in stable in their equilibrium positions where they have minimum potential energy. Once deflected from this position, the disturbances periodically propagate through the material in the form of waves. There are two modes of waves, which exhibit different properties in propagation: The longitudinal waves, where the atoms oscillate in the direction of propagation and cause periodic density fluctuations; and the transverse waves, where the atoms oscillate orthogonal to the direction of propagation. This is illustrated in fig. 1. Depending on the coupling of neighboring atoms that is mainly dependent on their bonding strength, these waveforms can propagate at different speed. In hard materials, for example Graphen, longitudinal waves propagate at a high speed of $c \approx 20000\,\text{m/s}$ [3] whereas in Gold, a rather soft metal, they only prop-

agate at velocity $c \approx 3240\,\mathrm{m/s}$ [4]. In general the sound velocity follows a material specific dispersion relation and thus depends on the frequency of the wave. These waves, which exhibit a discrete energy distribution, are also described in terms of particles that are called phonons.
In this experiment the propagation speed of ultrasound in Polyacryl and water was determined by using different piezoelectric crystals that oscillated at frequencies of 1 MHz and 2 MHz. To ensure the coupling between oscillator head and the different media, coupling gel was applied at the connected surface. All measurements were done by using the `GS-EchoView`, that showed every emitted and detected ultrasound signal.

3 Velocity of sound in different media

3.1 Measurement with polyacryl

To measure the propagation speed of sound waves through media, there were three polyacryl cylinders of different lengths. It was first measured the length with a caliper. The speed of sound was then determined with the method of reflection, where the ultrasound transmitter also worked as the detector. Because of the sound signal being reflected on the other end of the cylinder, the effective length that is traveled within a time interval of Δt is two times the length l of each cylinder. The velocity of sound is then calculated with:

$$c_{\text{poly}} = \frac{2l}{\Delta t} \tag{1}$$

This method was applied to measure the sound velocity for each length of these cylinders and for the frequencies of $f = 1\text{MHz}$ and $f = 2\text{MHz}$. For the error of our time measurement, it is chosen $\sigma_{\Delta t} = 0.2\,\mu\text{s}$. All relevant data is shown in table 1.

Within their range of error, the velocity values for $f = 1\text{MHz}$ are in agreement with each other. The same accounts for the frequency $f = 2\text{MHz}$. Therefore it is calculated the mean value for each frequency:

$$c_{\text{poly, 1MHz}} = (2626.2 \pm 6.5)\,\frac{\text{m}}{\text{s}}$$

$$c_{\text{poly, 2MHz}} = (2655.3 \pm 6.7)\,\frac{\text{m}}{\text{s}}$$

Now it is clear visible, that the speed of sound in polyacryl depends on the frequency of the wave that is traveling through it. Thus it indicates dispersion in sound velocity. In the data sheet from the manufacturer, it is given a longitudinal wave velocity of $\approx 2700\,\mathrm{m/s}$ for polyacryl. [5] Our obtained values do not match this literature value within their range of uncertainty. This is most probably due to the fact that the measurements was done in saturation of the used piezoelectric ultrasound emitter. But for all following experiments this mode wasn't changed at all and thus our findings were consistent despite working in saturation.
It should also be noted that with the 4 MHz ultrasound emitter we couldn't see any reflected signal and thus neglected the further measurements with this frequency. In section 5 it is tried to give an explanation for this observation.

l [mm]	f [MHz]	Δt [μs]	c_{poly} [m/s]
41.27 ± 0.01	1	31.5 ± 0.2	2629.8 ± 5.8
	2	31.1 ± 0.2	2653.0 ± 5.9
80.70 ± 0.01	1	61.4 ± 0.2	2628.7 ± 8.6
	2	60.7 ± 0.2	2659.0 ± 8.8
120.18 ± 0.01	1	91.4 ± 0.2	2620 ± 17
	2	90.6 ± 0.2	2654 ± 17

Table 1: Here are shown the sound velocity values that were measured with the reflection method and calculated with eq. (1). Within their range of uncertainty, every propagation speed that was measured for a single frequency is in agreement with the others. Comparing the velocity of different frequencies, it is evident that polyacryl exhibits dispersion.

3.2 Measurement with water

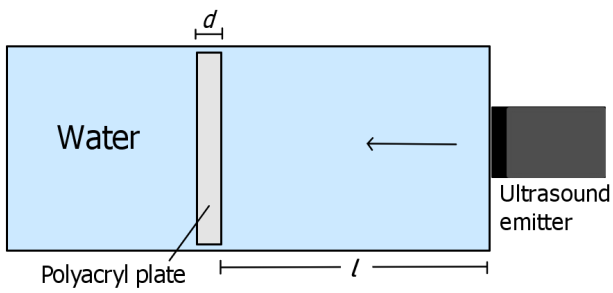

Figure 2: This figure illustrates the experimental setup for the measurement of sound velocity in water. It was used the reflection method with the 2 MHz ultrasound emitter and a plastic tube filled with water. As a reflector there was inserted a polyacryl plate of width d, which we placed at different distances l inside the water bath.

The same measurement was done using a plastic tube filled with water. Inside the tube, at distance l from the emitter was located a polyacryl plate of width d, as seem om fig. 2. It was measured the time interval Δt_1 of the emitted and detected signal for four different distances l. The speed of sound in

water then was evaluated with eq. (1). It should be noted that there was another reflected signal that has been detected with a time difference of Δt_2 to the emitted signal. This is because the ultrasound wave that is reflected from the front side of the polyacryl plate partly entered the medium and was reflected from the backside of the interior of the plate. After traveling the distance of $2d$ within polyacryl, the part of the signal that entered the small plate leaves it and travels back to the emitter. The time lapse of the two detected reflection signals is given as

$$\Delta\tau = \Delta t_2 - \Delta t_1 \tag{2}$$

By knowing the sound velocity of ultrasound with $f = 2\,\mathrm{MHz}$ inside polyacryl, which was determined in section 3.1, the width d of the polyacryl plate can be evaluated with:

$$d = \frac{\Delta\tau}{2} c_{\text{poly, 2MHz}} \tag{3}$$

The distance l was measured with a conventionally ruler. The error of reading was chosen to be $\sigma_l = 0,1\,\mathrm{cm}$ for each readout. Tabular 2 and 3 show the evaluated speed of ultrasound in water and the the width d that was determined by using eq. (2) and eq. (3).

l [cm]	$\Delta t_1 [\mu\mathrm{s}]$	c_{water} [m/s]
4.80 ± 0.14	64.5 ± 0.2	1488 ± 44
8.30 ± 0.14	112.7 ± 0.2	1473 ± 25
11.80 ± 0.14	161.0 ± 0.2	1465 ± 17
17.00 ± 0.14	228.7 ± 0.2	1487 ± 12

Table 2: This table consists of the measured time intervals Δt_1 for different distances l. The velocity of sound in water was calculated.

l [cm]	$\Delta t_2 [\mu\mathrm{s}]$	d [mm]
4.80 ± 0.14	72.0 ± 0.2	9.96 ± 0.38
8.30 ± 0.14	120.2 ± 0.2	9.96 ± 0.38
11.80 ± 0.14	168.4 ± 0.2	9.82 ± 0.38
17.00 ± 0.14	235.9 ± 0.2	9.55 ± 0.38

Table 3: This table shows the evaluated width d of the polyacryl plate with the additional needed time difference Δt_2, which originated from the second reflected signal.

For the final determination of the sound velocity in water for a ultrasound wave of frequency $f = 2\,\mathrm{MHz}$ it was calculated the mean value. We got:

$$c_{\text{water, 2MHz}} = (1478 \pm 14)\,\frac{\mathrm{m}}{\mathrm{s}}$$

Estimating that the used water had a temperature of $T \approx 20\,^\circ\mathrm{C}$, the literature value for the velocity of sound inside water with that temperature is given with $c_{\text{water}} = 1482.9\,\mathrm{m/s}$ [6]. Within the range of uncertainty, the determined velocity agrees with this literature value.
For each position of the polyacryl plate, the width d was calculated in tab. 3. The final width is also evaluated as the mean value:

$$d = (9.82 \pm 0.19)\,\mathrm{mm}$$

For comparison, the width of the polyacryl plate was additionally measured with a caliper after the experiment. We got $d_{\text{cal}} = (9.97 \pm 0.01)\,\mathrm{mm}$. Within its error range the determined width d, by considering both reflected signals, agrees with this guideline value.

4 Measurements on a polyacrylic test block

In this part of the experiment, it was placed a 1MHz ultrasound transmitter on the top side of a polyacryl-test block, which had regular bores of different size at various distance from its upper edge. A reference sketch is shown in fig. 3. From the top direction, ultrasound waves entered the polyacryl medium and were reflected back from each hole. The reflected signal was detected by the same echoscope. and it was measured the time interval Δt_{i} between emission and detection. The distance traveled within this time interval was exactly $2d_{\mathrm{i}}$. Knowing the previously determined speed of sound $c_{\text{poly, 1MHz}} = (2626.2 \pm 6.5)\,\mathrm{m/s}$ for $f = 1\mathrm{MHz}$ in polyacryl, it was evaluated the distance d_{i} at which the holes were located by applying the equation:

$$d_{\text{refl,i}} = \frac{\Delta t_{\mathrm{i}}}{2} c_{\text{poly,1MHz}} \tag{4}$$

To compare the determined distances of each borehole with a guideline value, it was additional measured by hand with a caliper. The error of reading is $\sigma_{\mathrm{d}} = 0.01\mathrm{mm}$. All the data is shown in tab. 4.

Within their range of uncertainty, the distance measurements with the sound reflection method agree with the obtained caliper values, except for the first two boreholes. This can be explained by considering the used saturation mode of the ultrasound emitter. The software `GS-EchoView` showed a broad emission signal. For example in fig.6 it is visible a comparable signal at $t \approx 0$, which is temporally extended. As a result, the reflected waves that traveled back from borehole 1 and 2 were partly detected within the decreasing slope of the emission signal and thus the extrapolation of Δt for these two holes is more of an estimation.

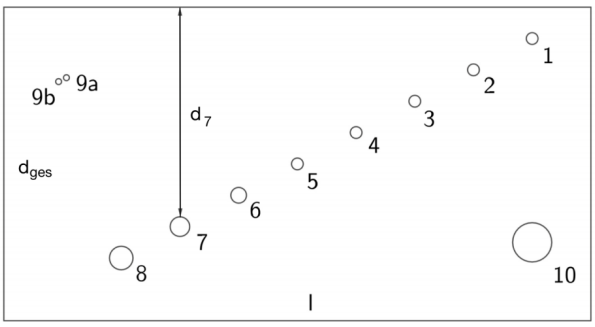

Figure 3: This sketch shows the polyacrylic test block which was used for the measurement. It has 11 bore holes of different size that are located in various distances d_i as seen above. Its length is $l = (150.02 \pm 0.01)$ mm.

i	Caliper d_i [mm]	Δt_i [μs]	$d_\mathrm{refl,i}$ [mm]
1	7.08 ± 0.01	4.3 ± 0.2	5.71 ± 0.26
2	15.10 ± 0.01	11.76 ± 0.2	15.44 ± 0.27
3	23.12 ± 0.01	17.73 ± 0.2	23.28 ± 0.27
4	31.11 ± 0.01	23.66 ± 0.2	31.07 ± 0.27
5	39.24 ± 0.01	30.01 ± 0.2	39.41 ± 0.28
6	46.58 ± 0.01	35.85 ± 0.2	47.08 ± 0.29
7	54.45 ± 0.01	41.47 ± 0.2	54.46 ± 0.30
8	61.65 ± 0.01	47.06 ± 0.2	61.80 ± 0.30
9a	17.46 ± 0.01	13.64 ± 0.2	17.67 ± 0.27
9b	19.41 ± 0.01	14.68 ± 0.2	19.27 ± 0.27
10	55.74 ± 0.01	42.42 ± 0.2	55.70 ± 0.30

Table 4: Distances of boreholes in the polyacrylic test block measured with reflection of ultrasound. For comparison it was added the data from manual caliper length measurements.

4.1 B-mode image

The software program `GS-EchoView` also offers a function that can record B-mode images, where the experimentalist has to move the ultrasound emitter with a constant velocity along a line on the surface of the probe. The result is a cross-sectional image representing the "reflective landscape" of the scanned object. It is constructed from echoes, which are generated by reflection of ultrasound on heterogeneous structures and irregularities inside the probe. Each echo is displayed at a point in the image, which corresponds to the relative position of its origin within the body cross section, resulting in a scaled map of echo-producing features. The brightness of the image at each point is related to the strength or amplitude of the echo, giving rise to the term B-mode (brightness mode). [1]
With the 1 MHz ultrasound source coupled to the surface represented by the line at the top in fig. 3, we recorded a gray scale B-mode image of the polyacryl brick. It is shown in fig. 4. In the appendix the same image is shown with a colored intensity scale bar (cf. fig. 7).

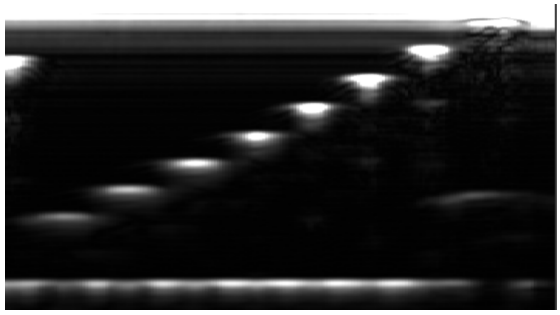

Figure 4: Gray scale B-mode image of a polyacrylic brick with several drillings. A 1 MHz ultrasonic emitter was used. The lighter a dot is, the higher the intensity of the reflected signal.

Several observations can be made in fig. 4: First, the bores $9a$ and $9b$ are too close for the signal to be able to resolve them. As discussed in chapter 6, this is because frequency of the sensor is too low. Hence for the 2 MHz, we would expect to be able to distinguish the bores in the B-mode image. Second, the farther away the reflection signature of a bore is located, the lower is its intensity. This is simply due to the damping of the medium polyacryl (cf. chapter 5). Third, a horizontal reflection signature can be seen at the bottom of the image. This line represents reflection at the surface opposite to the surface the scanner was coupled to. Fourth, the diameter of the bores varies: The scanning was done by hand which is why the speed of scanning could not be kept perfectly constant. The diameter of the bores was not constant either. Finally, the drilling 10 is not visible in the image because it is located "in the shadow of reflection" of drilling 1 and is thus not detectable by the scanner in this position.

5 Absorption of ultrasound in PVC

The coefficient of absorption is a material-specific quantity and indicates how far waves can travel in matter until its amplitude and thus its intensity is dampened below a certain threshold. The intensity of waves decreases exponentially with the distance Δx traveled in matter,

$$I \propto e^{-\Delta x/\lambda} \,. \tag{5}$$

Here λ is penetration depth that specifies how far the wave propagates until its intensity I decreased to $1/e$ of the initial value. With this definition the absorption coefficient ρ is defined as

$$\rho = \frac{1}{\lambda} = \ln\left(\frac{I(\Delta x)}{I(0)}\right)\frac{1}{\Delta x} \,. \tag{6}$$

In this part, we determined the coefficient of absorption of ultrasound in three PVC cylinders of the length $l_1 = (80.02 \pm 0.01)\,\mathrm{mm}$, $l_2 = (75.01 \pm 0.01)\,\mathrm{mm}$ and $l_3 = (72.63 \pm 0.01)\,\mathrm{mm}$, measured with a caliper. For this ultrasound emitters of the frequency $f_1 = 1\,\mathrm{MHz}$ and $f_2 = 2\,\mathrm{MHz}$ were coupled to the round area of each PVC cylinder with ultrasonic coupling gel. Then the cursor tool of the software program `GS-EchoView` was used to determine the amplitude A (the intensity is proportional to the square of the amplitude: $I \propto A^2$) of both the incident and reflected signal. We thus set $\Delta x = 2l_\mathrm{i}$ in eq. (6): The reflected signal has traveled twice the length of the cylinder when detected.
The resulting coefficients of absorption then were

$$\rho(f_1) = (0.093 \pm 0.017)\,\mathrm{cm}^{-1}$$

and

$$\rho(f_2) = (0.53 \pm 0.13)\,\mathrm{cm}^{-1}$$

The error of ρ is chosen to be the standard error of the series of measurement. Within this margin of error, the results for ρ do not agree, hence they are given separately. We thus conclude that the absorption depends on the frequency of the incident wave. The higher the frequency, the higher the absorption. This could explain why for the 4 MHz ultrasound source no reflected signal could be observed: Along the distance of its propagation, its amplitude decreased to values lower than the resolution of measurement.
It has to be added that the presented result must not be understood as exact but rather as qualitative: The precision of the cursor tool was unknown as well as the error of frequency of the emitters. Lastly, no information about the purity of the PVC was given.

6 Resolving power

In optics, two points are distinguishable, if the wavelength λ of the incident radiation is at most twice the distance d between these points: $\lambda \leq 2d$. This translates to: Two close neighboring points in solid material can only be resolved with ultrasound waves if the intensity of the detected signals, which were reflected from both points, drops below the half of their peak value in between these two peaks.
In this section we used the closely located drillings $9a$ and $9b$ of the polyacrylic test block from chapter 4, fig. 3 to evaluate the resolving power of the sound waves depending on their frequency. To this end it makes sense to use the full width at half maximum (FWHM) of a signal. Analogous to the definition of the resolution power in optics discussed above, the pulses reflected off the drillings are distinguishable only if

$$\mathrm{FWHM} \leq \Delta\tau, \tag{7}$$

where $\Delta\tau$ corresponds to the time difference of the separate reflected signals in adaption to eq. (2)
To measure the resolving power we logged the signal reflected by the bores $9a$ and $9b$ of the polyacrylic brick for sources with 1 MHz and 2 MHz. Then we used the cursor tool to determine the position, height and separation $\Delta\tau$ of both signal peaks as shown exemplarily for the 2 MHz source in fig. 5. Finally we computed the average FWHM by averaging the FHWM obtained separately for each peak. This had to be done since the FWHM for the second peak in fig. 5 was significantly greater than that of the first. We estimated the error of each read off position of the FWHM to $0.1\,\mu\mathrm{s}$; the error of the delta cursor-tool of the software `GS-EchoView` was estimated to $0.1\,\mu\mathrm{s}$. This step could not be done with higher precision because the software did not offer an option to export raw data. For $\Delta\tau$ we refrained from using the enveloping function (blue curve in fig. 5 and fig. 6 respectively) for its origin is unknown and it yielded different values than expected. Instead, we used the actual measured signal directly.

f/MHz	$\mathrm{FWHM}/\mu\mathrm{s}$	$\Delta\tau/\mu\mathrm{s}$
1	1.76 ± 0.14	1.29 ± 0.01
2	1.21 ± 0.14	1.27 ± 0.01

Table 5: The average full width at half maximum (AFWHM) of the reflection spectrum of two closely located bores in a polyacryl brick depending on the frequency of the ultrasound source. Only for $f = 2\,\mathrm{MHz}$ the peaks could be distinguished since $\mathrm{FWHM} < \Delta\tau$.

For 1 MHz the condition eq. (7) is not satisfied, hence the peaks can not be told apart. This is stringent with the qualitative observation that in fig. 6 in the appendix the intensity does not fall below half of its maximum value. For 2 MHz as depicted in fig. 5 however this situation can be observed. This was expected as for a smaller distance one needs a higher frequency to resolve it, similar to the statement of the Rayleigh criterion in optics.

This peak-to-peak method of measurement is legitimate only if done with constant frequency. Otherwise effects like dispersion widen the peaks and impede accuracy. It has to be noted that the 2 MHz emitter allowed a more precise read-off because its peaks at $U_\mathrm{max} \approx 0.83\,\mathrm{V}$ are better visible than those provided by the 1 MHz emitter at $U_\mathrm{max} \approx 0.50\,\mathrm{V}$. With the knowledge of the speed of sound in polyacryl c_poly (cf. chapter 3.1) the *horizontal* distance d_h between

Figure 5: Reflected spectrum of a 2 MHz ultrasonic pulse off two closely located bores in a polyacrylic brick. The yellow and green cursors are used to determine the temporal separation of the peaks. The horizontal cursors were not yet adjusted at the time the screen-shot was taken. Between the peaks it can be seen that the intensity drops below half of its maximum value, the pulses can thus be distinguished.

the bores $9a$ and $9b$ can be found with $d_\mathrm{h} = \Delta\tau c_\mathrm{poly}$:

$$d_\mathrm{h}^{1\,\mathrm{MHz}} = (1.691 \pm 0.013)\,\mathrm{mm}$$
$$d_\mathrm{h}^{2\,\mathrm{MHz}} = (1.686 \pm 0.015)\,\mathrm{mm}$$

The values match each other within the margin of error. The measurement of the vertical positions of the drillings with a caliper done in chapter 4 provides a value of reference by subtracting the vertical positions of bore $9a$ and $9b$:

$$d_\mathrm{h,ref.} = (1.680 \pm 0.14)\,\mathrm{mm}$$

This reference agrees with the obtained result as well. We thus conclude that if the speed of ultrasound inside a given medium is known, distances of ultrasound-reflecting objects can be determined with great precision.

7 Conclusions

In this experiment, it was used the reflection method to determine the speed of ultrasound in polyacryl for a frequency of $f = 1\,\mathrm{MHz}$ and $f = 2\,\mathrm{MHz}$. We got $c_\mathrm{poly,\,1MHz} = (2626.2 \pm 6.5)\,\mathrm{m/s}$ and $c_\mathrm{poly,\,2MHz} = (2655.3 \pm 6.7)\,\mathrm{m/s}$. This confirms that the medium consists of dispersion in sound. Further the velocity of ultrasound in water was determined with the 2 MHz emitter to be $c_\mathrm{water,\,2MHz} = (1478 \pm 14)\,\mathrm{m/s}$. This value agrees with the literature value within its range of uncertainty. The reflection method provided distance measurements of various boreholes in a polyacrylic test block, where the 1 MHz emitter was used. The results were compared with a caliper distance-measurement. Then, with the same frequency, a B-mode image of the test block was captured. On two close neighboring boreholes of this block, it was examined the resolution power of ultrasound waves for $f = 1\,\mathrm{MHz}$ and $f = 2\,\mathrm{MHz}$ by evaluating the full witdth at half maximum (FWHM). As a result, the resolution power was greater for the higher frequency. With three PVC-cylinders of different length, the absorption coefficient ρ was measured to be $\rho(f_1) = (0.093 \pm 0.017)\,1/\mathrm{cm}$ for $f = 1\,\mathrm{MHz}$ and $\rho(f_2) = (0.53 \pm 0.13)\,1/\mathrm{cm}$ for $f = 2\,\mathrm{MHz}$. This gives evidence that higher frequencies are more damped inside PVC.

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8 Appendix

Figure 6: Reflected spectrum of a 1 MHz ultrasonic pulse off two closely located bores in a polyacrylic brick. Between the peaks the measured voltage (and thus the intensity) does not drop below its peak value, thus the peaks are not distinguishable.

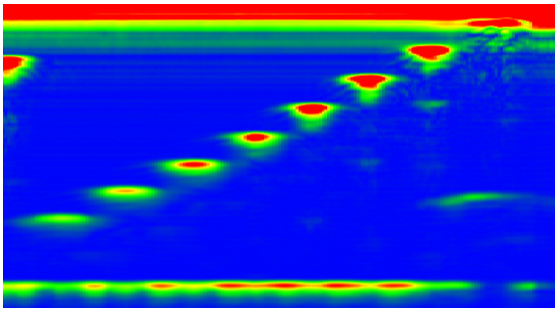

Figure 7: B-mode image of a polyacrylic brick with several boreholes. A 1 MHz ultrasonic emitter was used.